

Bennu: Recipe For Life [B1]

La NASA sta studiando i campioni prelevati da questo asteroide dal nome egizio: contengono importanti informazioni sulla formazione del sistema solare e sull'origine della vita sulla Terra.



Millions of miles from Earth, a robotic [probe](#) collected [samples](#) from an asteroid named Bennu in 2018. Now NASA scientists are studying those [samples](#), which are suggesting new possibilities about the origin of life.

AS OLD AS TIME

Asteroid Bennu is named after the ancient Egyptian mythological bird Bennu, a symbol of the Sun, creation, and rebirth. It is about five hundred meters wide and is technically a [rubble pile](#) asteroid. This means that it's not one solid rock but a collection of rocks and [debris held together](#) by gravity and electrostatic forces. These rocks and [debris](#) were originally part of a much larger asteroid that formed in the outer Solar System about 4.6 billion years ago — when our Solar System [came into being](#).

INGREDIENTS FOR LIFE

Bennu was discovered by NASA in 1999, as part of a project to find and [track](#) asteroids and comets near Earth; 'near', of course, is relative in this case, as the asteroid was about 7.4 million kilometres from our planet. In 2016, NASA launched a robotic [probe](#) named OSIRIS-REx. It arrived on Bennu in 2018 and collected [samples](#) from it. Those [samples](#) then returned to Earth inside a capsule, when they [parachuted down](#) to a desert in the US state of Utah in 2023. Places

Greenwich: The Centre Of Time And Space È qui che inizia ogni giornata sulla Terra: a Greenwich, il sobborgo di Londra da cui passa il meridiano zero. Una convenzione stabilita nel lontano 1884. Justin Ratcliffe Since then, NASA scientists have been studying the [samples](#). They have found that the asteroid contains 16,000 types of organic molecules, including all the most important components needed for life: amino acids, essential sugars, carbon-rich molecules [and so on](#). This suggests the possibility that asteroids formed in space at the beginning of our Solar System may have transported these components to Earth billions of years ago, so [seeding](#) life on our planet.

ECOSYSTEM COLLAPSE

Scientists around the world have commented on the importance of NASA's analysis of the asteroid. Mark Schneegurt, an astrobiologist at Wichita State University, told The New York Times: "There could hardly be any study more important to our understanding of the origins of life in the Solar System." However, there is a possibility that the asteroid that is giving us crucial information about the origin of life on our planet could one day destroy it. NASA has said that there is a small chance that Bennu will impact Earth in the future,

causing ecosystem collapse, with the highest probability of impact on 24 September 2182.

Glossary

- **rubble pile** = cumulo di detriti
- **held together** = mantenersi unito
- **track** = tracciare
- **and so on** = e così via
- **probe** = sonda spaziale
- **debris** = detriti, macerie
- **came into being** = nascere, formarsi
- **parachuted down** = scendere con il paracadute
- **seeding** = piantare, seminare
- **samples** = campioni